

Gangadhar K

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EXPERIENCE

AI / ML Engineer

CRETechHub

October 2024 - Present, Hyderabad

- Applied machine learning techniques for anomaly detection and forecasting to analyze and predict trends in complex databases.
- Developed an interactive agent-based chat application with FastAPI endpoints.

Generative AI Engineer

Zerzura Studios Private Ltd

September 2024 - Present

- Developed diffusion models for generating and modifying game character sprites using text and image prompts.
- Led a team and developed an AI-powered educational kid's app, integrating TTS and STT for an interactive learning experience.

Secondary Investigator

Innovation, Incubation, and Entrepreneurship Cell (IIEC) VIT-AP

July 2023 - Present, Amaravathi

- Defined project scope, timeline, and budget for the development of an autonomous VTOL vehicle.
- Collaborated with a team of seven student engineers to design, build, and test a functional VTOL prototype.
- Developed and implemented computer vision-based applications using the VTOL vehicle.

AI/ML Researcher

DigitalFortress Private Limited

July 2023 - December 2023, Amaravathi, AP

- Actively involved in Chatbot development, seamlessly integrating OpenAI API Key, MongoDB database and LangChain for comprehensive solution.
- Successfully implemented a facial recognition model using a Siamese network approach, demonstrating proficiency in diverse technologies and frameworks.

Full Stack Developer Intern

Sreejay Technocreations Pvt. Ltd

September 2022 - December 2022, Hyderabad

- Involved in the development of IoT & AI enabled product, SmartEdificio as a full stack developer in the Transform offering of SmartEdificio which is focused on smartening property and facility management services in commercial real estate.
- Completed certification in React, Python and Django. Developed screens and associated services.

PROJECTS

What-If - Explainable AI Image Counterfactual Generator

November 2024 - Present

- Developed a ViT-based classifier with explainable AI to generate and visualize counterfactual explanations for image predictions.
- Enabled interactive, real-time predictions, class selection, and counterfactual generation in a Streamlit app.
- Visualized key classification changes with heatmaps, confidence metrics, and pixel change analysis.

ENVIO - AI Based Package and Environment Manager

October 2024 - Present

- Used multiple CrewAI Agents and LLM's to create an interactive, personalized AI to aid users in creating python environments and manage package dependencies (Capstone Project).

Comparative Analysis of Vision Models on Underground Mining Classification

October 2024 - November 2024

- Achieved a 4-5% increase in mAP50 accuracy (from an initial 80-90%) for object detection in underground mining scenarios by optimizing deep learning models (YOLOv8-v12, Faster R-CNN, DETR). Implemented data augmentation and model refinement techniques, focusing on real-time detection of coal miners, coal chunks, and safety-related elements.

Monocular Depth Estimation using UNET Architecture

May 2023 - August 2023

- Utilized the UNET architecture to predict depth from single images.
- Implemented custom accuracy and loss functions to optimize the model's performance.
- Leveraged the NYU Depth V2 dataset for training the model.
- Achieved accurate depth predictions applicable to fields such as scene understanding, robotics, and augmented reality.

CRISP - Comprehensive Route Information System for Passengers

January 2022 - May 2022

- Developed a dynamic bus tracking solution using Arduino, Spring Boot with Java, and Flutter for a user-friendly interface, enabling real-time

- route tracking.
- Built CRISP, a React-based admin panel for efficient management and oversight of university bus routes, enhancing passenger experience.

PUBLICATIONS

CRISP - Comprehensive Route Information System for Passengers

A modular, dynamic bus tracking solution using Arduino, Spring Boot with Java, and a user-friendly mobile interface created with Flutter. 9th National Conference on Advancements in Information Technology (NCAIT-2023), JSS Academy of Technical Education (May 3, 2023).

Comparative Analysis of Vision Models on Underground Mining Classification

A comparative analysis of Vision models for underground mine classification, evaluating their performance and accuracy in detecting and classifying objects in mining environments. (Submitted to Artificial Intelligence Review)

Analysis of Social Media Sentiments in the Russia-Ukraine Conflict using NLP and ML Methods

This work provides a comprehensive sentiment analysis of tweets regarding Russia-Ukraine conflict using Natural language processing techniques. (Submitted to Scientific Reports)

Monocular Depth Estimation using UNET Architecture

This publication explores using Monocular depth estimation model that estimates object depth from a single image, colorizing it to visually represent spatial relationships. (To be submitted)

EcoSync: Decentralized Global Economic Forecasting Using Federated Learning

This publication explores using federated learning to build a global economic forecasting model that preserves data privacy by training locally on sensitive economic data from multiple regions. (To be submitted)

What-If - Explainable AI with Counterfactual Explanations for Vision Transformers

This paper investigates the use of explainable AI techniques, specifically counterfactual explanations, with Vision Transformers in image classification, offering transparent insights into model decisions and improving interpretability. (To be submitted)

SKILLS

Languages: Python, HTML, CSS, JS, SQL

Technologies: Linux, Git, BitBucket

Frameworks/Tools: TensorFlow, Pytorch, OpenCV, Scikit-learn, Keras, NLTK, CrewAI, HuggingFace, Langchain, Streamlit, Agents, FastAPI.

Cloud Services: Azure, Docker, AWS

Tech Domains: Deep Learning, Computer Vision, Natural Language Processing, Statistical Machine Learning, Information retrieval, RAG, Federated Learning, Explainable AI

CERTIFICATIONS

- **AWS Cloud Practitioner**
Amazon Web Services (AWS)
Issued: [Jan, 2023]

EDUCATION

Bachelor of Technology in Computer Science and Engineering, specializing in Artificial Intelligence and Machine Learning

Vellore Institute of Technology - AP • Amaravathi, AP • 2025 • 8.6

Higher Secondary Education

SRI CHAITANYA JUNIOR KALASALA • Hyderabad • 2020 • 94%